

STD – X
Subject : Science



CHAPTER 1
**Chemical Reactions
and Equations**





CHAPTER 1

Chemical Reactions and Equations

Part – 8
Corrosion and Rancidity

Corrosion



EFFECTS OF OXIDATION REACTION IN EVERY DAY LIFE

Rusting

- Iron articles are shiny when new, but get coated with a reddish brown powder when left for some time. This process is commonly known as rusting of iron.



Corrosion

- The process in which a metal is attacked by substances around it such as moisture, acids, etc., it is called corrosion.
- Corrosion takes place due to the oxidation reaction.
- Corrosion cause damage to metal objects such as car bodies, bridges, iron railings, ships etc.

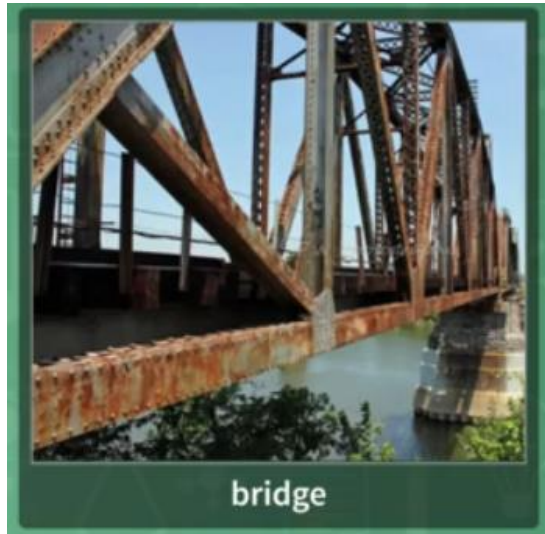
EFFECTS OF OXIDATION REACTION IN EVERY DAY LIFE

Corrosion

- Corrosion cause damage to metal objects such as car bodies, bridges, iron railings, ships etc.
- The reddish brown powdery substance is called rust.



Car Bodies



bridge



Iron Railing



Ship

EFFECTS OF OXIDATION REACTION IN EVERY DAY LIFE

- **Corrosion :**

- **Examples of corrosion**

- 1) Iron articles get coated with a reddish brown powder when left exposed to moisture in rainy days. This process is commonly called as 'rusting' of iron.
- 2) Copper objects gets corroded with green coating.
- 3) Silver articles gets corroded by black coating.



Methods of prevention of corrosion.



Keeping Iron objects in dry place.



Coating with oil paint.



Applying lubricating oil or grease.



Using techniques like Anodizing

Anodizing is a process that uses electrolysis to create a thick oxide layer on a metal surface. It's often used on aluminum and its alloys.



Alloying

Alloying is the process of combining two or more metals or elements to create a new substance. Alloys are used in many engineering applications because they have better properties than pure metals

Rancidity



EFFECTS OF OXIDATION REACTION IN EVERY DAY LIFE

- Rancidity :

[VIDEO - 1.3.2 - \(RANCIDITY\).mp4](#)

- The process in which fat or oil becomes rancid and they get undesirable taste and smell is called rancidity.
- Rancidity is also caused due to oxidation reaction of fat and oil.



Suggest some methods of preventing rancidity.



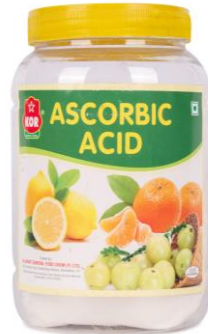
Flushing the bags of Chips with nitrogen gas to prevent the chips from oxidation.



Keeping food at low temperature in refrigerators.



Keeping food in air tight containers.



Substances which prevent oxidation (antioxidants) are added to foods containing fats and oil.



Home - Work



Q U E S T I O N S

1. Why does the colour of copper sulphate solution change when an iron nail is dipped in it?
2. Give an example of a double displacement reaction other than the one given in Activity 1.10.
3. Identify the substances that are oxidised and the substances that are reduced in the following reactions.
 - (i) $4\text{Na(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{Na}_2\text{O(s)}$
 - (ii) $\text{CuO(s)} + \text{H}_2\text{(g)} \rightarrow \text{Cu(s)} + \text{H}_2\text{O(l)}$





Home - Work



Q U E S T I O N S

1. Why does the colour of copper sulphate solution change when an iron nail is dipped in it?

Ans : 1) When an iron nail is dipped in copper sulphate solution, iron being more reactive than copper, it displaces copper from the solution.

2) As a result, iron sulphate is formed, which is green in colour.

3) Thus, colour of copper sulphate solution changes.





Home - Work



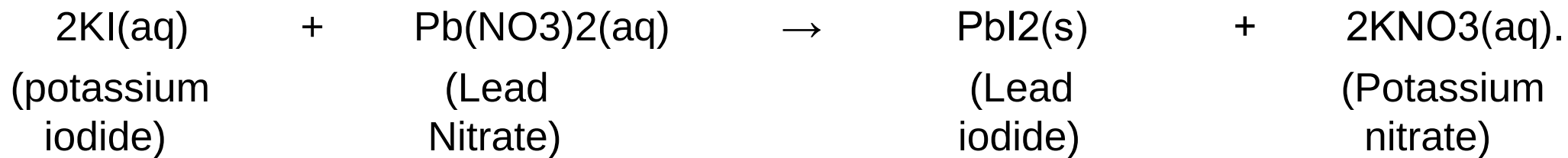
2. Give an example of a double displacement reaction other than the one given in Activity 1.10.

Ans:1) The reaction between potassium iodide (KI) and lead nitrate ($\text{Pb}(\text{NO}_3)_2$) is a double displacement reaction.

2) In this type of reaction, two compounds exchange ions to form two new compounds.

3) In this reaction, lead iodide (PbI_2) is formed as a yellow precipitate and potassium nitrate (KNO_3) as an aqueous solution.

4) The reaction is represented by the following balanced chemical equation:





Home - Work



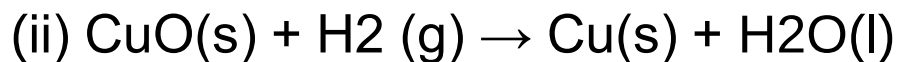
3. Identify the substances that are oxidised and the substances that are reduced in the following reactions.



Ans : (i) $4\text{Na(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{Na}_2\text{O(s)}$

In this reaction,

- Sodium is oxidised, because it sodium gains oxygen (and loses electrons.).
- Oxygen is reduced, because oxygen gains electrons.



In this reaction,

- CuO is reduced, because it loses oxygen (and gains electrons).
- H₂ is oxidised, because it gains oxygen (and loses electrons).

THANK YOU

